



Research Article

Explainable ensemble machine learning for financial transaction fraud detection: Insights from XGBoost and deep neural networks

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ARTICLE INFO

Keywords:

Fraud detection
Machine learning
XGBoost
Deep learning
Ensemble models

ABSTRACT

The rapid digitalization of financial services has enhanced transaction speed and accessibility but also amplified exposure to fraud activities that undermine institutional integrity and consumer trust. Effective fraud detection requires analytical frameworks that are not only accurate and adaptive but also interpretable and compliant with financial regulations. This study develops a scalable, explainable machine learning framework for detecting fraud financial transactions using a hybrid ensemble of Extreme Gradient Boosting (XGBoost) and Deep Neural Networks (DNN). The proposed model integrates transactional, temporal, and identity-linked features to capture behavioral and contextual patterns in large-scale, high-frequency data. To address the extreme class imbalance inherent in fraud detection, we evaluate multiple strategies including focal loss, class weighting, and threshold optimization. Model transparency is enhanced through SHAP-based interpretability analysis, providing granular insights into the feature interactions driving fraud risk. Empirical evaluation on a real-world transaction dataset demonstrates that the hybrid ensemble achieves superior detection accuracy and recall relative to baseline models while maintaining explainability suitable for regulated financial environments. The results highlight the potential of combining interpretable machine learning with adaptive ensemble learning to enhance resilience and trustworthiness in modern financial risk management systems.

1. Introduction

The rapid digitalization of financial services (Broby, 2021) has fundamentally transformed the way individuals and businesses conduct transactions across e-commerce, mobile payments, and on-line banking. While these innovations have enabled unprecedented convenience and efficiency, they have also introduced new vulnerabilities, exposing financial systems to increasingly sophisticated fraud activities. Credit card fraud, in particular, remains one of the most pervasive and costly financial crimes, causing direct monetary losses and eroding public trust in financial institutions (Ding et al., 2025). Accurately identifying fraud is challenging because malicious activities are often concealed within massive streams of legitimate transactions, making traditional detection approaches insufficient.

Early fraud detection systems were predominantly rule-based, relying on static thresholds and handcrafted anomaly detection rules to flag suspicious transactions (Bolton and Hand, 2002). Although such approaches were initially effective, they fail to adapt to

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Peer review under the responsibility of KeAi Communications Co., Ltd.

<https://doi.org/10.1016/j.jfds.2026.100195>

Received 3 November 2025; Received in revised form 29 May 2026; Accepted 2 June 2026

Available online 9 June 2026

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evolving fraud patterns. Fraudsters continuously modify their tactics, rendering fixed rules obsolete and leading to high false-positive rates while allowing genuine fraud cases to go undetected (Ding et al., 2025). These limitations have driven a paradigm shift toward machine learning (ML) and deep learning (DL), which can automatically learn complex, non-linear patterns from data, adapt to emerging fraud tactics, and uncover relationships that are difficult to model manually (Roseline et al., 2022).

ML has shown particular strength in analyzing large-scale, high-dimensional transaction data (Talukder et al., 2024; Zhang et al., 2019). Cost-sensitive learning has been proposed to explicitly account for the financial consequences of misclassification, striking a balance between minimizing false positives and maximizing fraud detection (Roseline et al., 2022). To address scalability requirements, distributed learning frameworks have been developed to process high-velocity data streams in near real time (Theodorakopoulos et al., 2025). In addition, various algorithms have been explored: support vector machines and neural networks effectively capture non-linear relationships, whereas gradient boosting methods such as XGBoost offer robust performance and strong predictive accuracy for tabular, structured data (Noviandy et al., 2023).

Despite these advances, fraud detection still faces several challenges. One of the most persistent is the extreme class imbalance between legitimate and fraud transactions, where fraud often accounts for less than 1% of all records (Breskuvienė and Dzemyda, 2024). This imbalance biases models toward the majority class and reduces recall for fraud cases, which is particularly costly in real-world applications. Existing solutions include active learning to prioritize informative samples (Riskiyadi, 2024), ensemble boosting methods tailored to minority classes, and resampling strategies such as SMOTE and undersampling to rebalance the data distribution (Ludera, 2021).

Another major challenge is the high dimensionality and structural complexity of financial transaction data, which involve non-linear feature interactions. Feature engineering has therefore become central to improving model performance, with techniques such as time-window aggregation to capture sequential spending behavior (Saputra et al., 2019). Recent research highlights the promise of hybrid ML–DL approaches, which combine the predictive power of deep neural networks with the interpretability of tree-based models (Gandhar et al., 2024). Advanced feature selection frameworks that integrate statistical and evolutionary methods have also been proposed to reduce redundant features, lower computational cost, but retain discriminative power (Siam et al., 2025).

DL has emerged as a dominant paradigm in recent years due to its ability to automatically extract high-level representations from raw transaction data (Du et al., 2023; Zioviris et al., 2024). Deep neural networks have been shown to scale well to distributed systems and maintain generalization in dynamic environments (Prasad and Srikanth, 2024). Furthermore, their ability to perform anomaly detection and representation learning makes them highly effective when fraud patterns evolve over time (Mehbodniya et al., 2021). Nevertheless, DL models often require large volumes of labeled data, demand substantial computational resources, and suffer from limited interpretability, posing challenges for deployment in highly regulated financial environments (Mienye et al., 2024).

Building on these insights, the present study addresses these gaps through a comprehensive investigation of supervised ML approaches for fraud detection of financial transactions. Specifically, the study (i) systematically compares seven ML models, including logistic regression, random forests, XGBoost, shallow and deep neural networks, and a novel hybrid ensemble; (ii) evaluates imbalance handling strategies, such as class weighting, focal loss, and threshold optimization; (iii) incorporates transactional, temporal, and identity-linked features to improve contextual predictive power; and (iv) applies SHAP-based interpretability analysis to ensure transparency and regulatory compliance. The main contributions of this research are threefold: (1) a detailed characterization of fraud patterns across transactional, temporal, and identity dimensions; (2) a hybrid ensemble model integrating Enhanced XGBoost and a Deep Neural Network, achieving strong predictive performance on the dataset we consider; and (3) an adaptive evaluation framework that integrates model interpretability and dynamic threshold optimization to support practical application in fraud detection systems.

2. Related work

Research on fraud detection has largely focused on supervised ML approaches, aiming to address the challenges posed by real-world datasets that are highly imbalanced, high-dimensional, complex and often noisy. Existing studies can be broadly classified into four themes: (i) model selection, (ii) imbalance-handling techniques, (iii) feature engineering, and (iv) the integration of identity-based variables to improve predictive power of the model.

Supervised learning models remain fundamental for most modern fraud detection systems (Bin Sulaiman, Schetinin and Sant, 2022; George et al., 2025). These models learn parameters or hyper-parameters from labeled historical data and build the model to predict whether new transactions are fraud or legitimate. A notable advancement in this area is the distributed deep neural network (DDNN) proposed by Lei et al. (2023), which balances predictive accuracy with user privacy by allowing institutions to train local models and share only model parameters with a central server. This federated-style approach not only preserves data confidentiality but also improves efficiency through distributed computation, yielding superior accuracy, precision, recall, and F1-scores compared to centralized models (Xia and Saha, 2025).

Interpretable models such as decision trees and random forests have also been widely used in fraud detection (Lee et al., 2025; Sun, 2025; Wajji et al., 2024). Decision trees are valued for their transparency but are prone to overfitting, whereas random forests mitigate this issue through tree aggregation and hyperparameter tuning (Shah and Sharma, 2023). However, both approaches are sensitive to severe class imbalance. A comprehensive evaluation of 66 algorithm–resampling combinations by Alfaiz and Fati (2022) highlighted that pairing strong classifiers such as CatBoost with effective undersampling methods like AllKNN can achieve state-of-the-art performance in terms of F1-score, recall, and AUC.

Class imbalance remains one of the most critical obstacles in fraud detection (Baisholan et al., 2025; Velarde et al., 2023). When fraud cases account for less than 1% of transactions, models are significantly biased toward the majority class, leading to poor recall. Numerous resampling and reweighting techniques have been proposed to mitigate this challenge. For example, Alamri and Ykhlef

(2024) introduced BCB-SMOTE, a hybrid method combining Tomek links, clustering, and borderline synthetic oversampling, which achieved an F1-score of 85.2% while reducing overlap between classes. Similarly, [Ileberi et al. \(2021\)](#) showed that combining SMOTE with AdaBoost improved detection rates across classifiers, and [Lahbiss and Chtouki \(2024\)](#) reported that SMOTE-ENN with advanced models such as Random Forest and Long Short Term Memory (LSTM) significantly enhanced AUC-ROC. Moreover, [Jiao et al. \(2022\)](#) proposed DES-ICD, which integrates adaptive oversampling (AnnSMOTE) with dynamic ensemble selection to handle both class imbalance and concept drift. By generating minority samples that reflect new concepts and selecting classifiers based on local neighborhood performance, DES-ICD achieved superior accuracy and recall across multiple real and synthetic datasets. These findings highlight that careful design of imbalance handling techniques is crucial for achieving high recall without sacrificing precision.

Beyond class imbalance, feature engineering plays a pivotal role in fraud detection ([Alamri and Ykhlef, 2024](#); [Sharma et al., 2025](#)). The inclusion of behavioral and identity-based attributes has been shown to significantly improve model performance. For instance, [Shimin et al. \(2020\)](#) demonstrated that combining financial transaction data with identity-linked features (e.g., device type, email domain) significantly enhanced XGBoost's ROC-AUC to 0.942 on the IEEE-CIS dataset. Similarly, [Lucas et al. \(2019\)](#) applied Hidden Markov Models to capture sequential spending behavior, which, when combined with Random Forest, improved precision–recall AUC. [Bahnsen et al. \(2016\)](#) further extended this line of work by modeling periodic spending patterns with von Mises distributions, yielding a 13% reduction in financial losses. These studies collectively underscore that well-designed feature engineering strategies are essential for building robust and discriminative fraud detection models.

Ensemble and hybrid learning methods represent a growing trend in fraud detection research. [Carcillo et al. \(2021\)](#) combined unsupervised anomaly detection with supervised learning, feeding outlier scores as features into classifiers to enable multi-level detection. Dynamic ensemble selection approaches, such as the one proposed by [Achakzai and Peng \(2023\)](#), adaptively choose classifiers based on local competence and consistently outperform static ensembles. Hybrid models have also demonstrated promise; for example, [Jahnavi et al. \(2024\)](#) combined decision trees with logistic regression to achieve 98.1% accuracy for the data they considered, while [Chaurasia, Kesharwani, Sharma, Sharma, and Chugh \(2024\)](#) confirmed that XGBoost paired with data balancing strategies offers superior recall in rare-event detection.

Several comparative reviews have synthesized findings across models and datasets. [Patel et al. \(2024\)](#) reported that while deep neural networks often achieve the highest accuracy (up to 98.9%), simpler models such as logistic regression and Naive Bayes remain competitive due to their interpretability and high efficiency. Similarly, [Bhardwaj et al. \(2024\)](#) found that deep neural networks trained using the Adam optimizer reached 99.4% accuracy on the European credit card dataset and were computationally efficient, making them well-suited for large-scale data implementation.

Taken together, prior work demonstrates that deep learning and advanced ensemble approaches frequently deliver superior predictive performance, yet simpler models retain value for their interpretability, scalability, and ease of application. Despite these advances, few studies have simultaneously addressed all major challenges, including class imbalance, feature selection, interpretability, and integration of identity-based features, within a unified framework. This work seeks to close this gap by systematically comparing multiple supervised ML models, integrating advanced imbalance handling techniques, and leveraging feature engineering to deliver a scalable, interpretable, and robust fraud detection framework.

3. Materials and methods

This study adopts a systematic and rigorous methodology for evaluating supervised machine learning models in the context of financial fraud detection. The proposed framework is explicitly designed to address two major challenges inherent to large-scale financial transaction data: high dimensionality and severe class imbalance. The methodological design is organized into four sequential phases, data preprocessing, imbalance mitigation, model training with hyperparameter optimization, and post-hoc interpretability analysis. Each phase is implemented to ensure both model robustness and transparency. The following sections provide detailed descriptions of the procedures and techniques applied within each phase.

3.1. Data and its description

The data used in this study is the IEEE-CIS fraud detection dataset, released through a Kaggle competition in collaboration with Vesta, a global fraud prevention company. This data reflects real-world e-commerce environments with anonymized transaction and identity information, and is very suited for testing algorithms and computational framework designed for fraud detection in financial transactions. The training set contains over 590,000 records, of which only about 3.5% are labeled as fraud, while the test set contains similar features but does not include fraud labels. The dataset can be downloaded from the official Kaggle competition portal (<https://www.kaggle.com/competitions/ieee-fraud-detection/data>).

Two main files were provided for both training and test: transaction.csv, which contains transaction-level attributes, and identity.csv, which includes device and identity-related variables. These files were merged using the TransactionID field to produce a comprehensive view of each transaction. The features can be grouped into several categories, as summarized in [Table 1](#).

Beyond the feature types, it is also important to consider the overall statistical profile of the dataset. Basic descriptive statistics are provided in [Table 2](#). The dataset is highly imbalanced, with fraud cases representing only a small fraction of transactions. Transaction amounts vary widely, ranging from a few cents to over \$10,000, with a median value of approximately \$68. Additionally, many identity-related fields contain substantial missing values, highlighting the challenges inherent in real-world fraud detection problems.

These characteristics emphasize the dual challenges of extreme class imbalance and data high-dimensionality. These insights directly informed the preprocessing strategies and modeling decisions described in the following sections.

Table 1
Summary of feature groups in the IEEE-CIS dataset.

Feature Group	Description/Examples
Transaction features	TransactionAmt, TransactionDT, ProductCD
Card attributes	card1-card6 (e.g., card type, issuer, category)
Address codes	addr1, addr2 (geographic location codes)
Engineered features	C1-C14, D1-D15, V1-V339 (anonymized signals)
Email domains	P_emaildomain, R_emaildomain
Identity features	DeviceType, DeviceInfo, id12-id38 (browser, OS, network)

3.2. Data preprocessing

The training dataset was constructed by merging transaction dataset and identity dataset on the TransactionID field, thereby integrating transaction-level payment attributes with device- and identity-related features. This merging step was crucial to capture both behavioral and contextual features that can distinguish fraud from legitimate transactions. To address missing data, categorical variables were imputed with the string “missing” so that models could treat absence of information as an additional informative category. For numerical attributes, two different strategies were adopted depending on the model type. For tree-based methods such as XGBoost and Random Forest, missing values were retained as NaN, allowing these models to handle missingness natively during split optimization.

Since neural network models (both ANN and DNN) cannot process missing values directly, for all neural network-based models, numerical features with missing values were imputed using the median of each feature computed from the training set. Median imputation was chosen due to its robustness to skewed distributions and extreme values, which are common in financial transaction data. After imputation, numerical features were standardized using z-score normalization to ensure stable and efficient training of the neural networks. This hybrid preprocessing strategy ensures that each model type operates under conditions best suited to its underlying assumptions while maintaining consistency across the experimental pipeline.

Categorical features, including ProductCD, card4, and DeviceType, were transformed using label encoding to convert string categories into numerical form while preserving their distinct identities. Although more sophisticated encoders (e.g., target or one-hot encoding) could be applied, label encoding was selected to maintain consistency across a high-dimensional feature space and reduce memory overhead. In addition, a temporal feature, hour_of_day, was derived from the continuous TransactionDT timestamp to capture periodic spending behaviors that may indicate fraud, such as late-night or off-hour activity.

Following data preprocessing, the dataset was partitioned into training and validation subsets using an 80/20 stratified split. Stratification ensured that the proportion of fraud to legitimate transactions was maintained in both sets, enabling a fair and representative evaluation. The computational details of this preprocessing and splitting are presented in Algorithm 1. This approach preserved the natural distribution of the data, which is essential for imbalanced classification problems. However, it also retained potential noise from weak or redundant features, meaning that subsequent feature selection and model regularization were critical for improving robustness.

Algorithm 1: Preprocessing and Stratified Split

Input: Transaction file T , Identity file I
Output: $(X_{train}, y_{train}, X_{val}, y_{val})$
 Load T and I from CSV files;
 Replace hyphens in column names with underscores;
for each categorical column c in T and I do
 | Fill missing values in c with “missing”;
 Merge T and I on TransactionID;
for each categorical column c in merged dataset do
 | Apply Label Encoding to c ;
if TransactionDT exists then
 | Derive new feature hour_of_day $\leftarrow$ (TransactionDT//3600) mod 24;
 $X \leftarrow$ all features except isFraud, TransactionID;
 $y \leftarrow$ isFraud;
 Split (X, y) into (X_{train}, y_{train}) and (X_{val}, y_{val}) with 80/20 stratified split;
return $(X_{train}, y_{train}, X_{val}, y_{val})$;

Table 2
Descriptive statistics of the IEEE-CIS training dataset.

Statistic	Value
Total records	~590,000
Fraud proportion	3.5%
Transaction amount range	\$0.01 – \$10,000+
Median transaction amount	\$68
Number of features (after merge)	434

3.3. Handling class imbalance

Fraud transactions represented only about 3.5% of the dataset, creating a severe class imbalance that posed a significant challenge for model training. If left unaddressed, most classifiers would become biased toward predicting the majority class (legitimate transactions), thereby achieving deceptively high accuracy but failing to identify the rare fraud cases that matter most in practice. To mitigate this imbalance, cost-sensitive learning approaches were adopted. For gradient boosting models such as XGBoost, the parameter value of `scale_pos_weight` was set to the ratio of majority to minority class instances, thereby instructing the algorithm to assign higher importance to fraud cases during training. For neural networks, a focal loss function was used. Unlike traditional cross-entropy, focal loss dynamically down-weights well-classified examples and focuses learning on harder, misclassified fraud cases. This adaptation is particularly effective when fraud behavior exhibits high diversity and overlaps with legitimate transaction patterns. This strategy preserved the natural class distribution of the dataset and avoided the introduction of synthetic artifacts, which are often a drawback of oversampling or SMOTE-based techniques. However, while cost-sensitive methods improve recall, they may not fully resolve imbalance in scenarios where the decision boundary between classes is highly non-linear or overlapping.

3.4. Machine learning models

In this section, we provide a brief overview of the machine learning models used in this study to ensure the paper remains self-contained. Five models were selected based on their suitability for handling structured, imbalanced data classification problems and their diversity in algorithmic approach. Let $\{\mathbf{x}_i, y_i\}_{i=1}^n$ denote a training set of financial transaction data, where $\mathbf{x}_i \in \mathbb{R}^p$ is the p -dimensional feature vector, and $y_i \in \{0, 1\}$ is the class label of fraud or non-fraud. We first consider Logistic Regression (LR), a linear baseline model that estimates the conditional probability of the positive (fraud) class as

$$\hat{y}_i = P(y_i = 1 | \mathbf{x}_i) = \sigma(\mathbf{w}^\top \mathbf{x}_i + b), \quad (1)$$

where $\sigma(z) = \frac{1}{1+e^{-z}}$ is the logistic sigmoid function, and $\hat{y}_i \in [0, 1]$ represents the predicted probability of fraud. The unknown parameters $\mathbf{w}$ and b can be estimated using the least squares method or maximum likelihood estimation.

To capture non-linear patterns, we also apply a Random Forest (RF) model, which is an ensemble of T decision trees $\{h_t\}_{t=1}^T$. Each tree is trained on a bootstrap sample of the data with random feature subsampling. The final prediction is the average of individual tree predictions of probability,

$$\hat{y}_i = \frac{1}{T} \sum_{t=1}^T h_t(\mathbf{x}_i), \quad (2)$$

where $h_t(\mathbf{x}_i) \in [0, 1]$ denotes the probability assigned by tree t .

As a more powerful alternative, XGBoost uses gradient boosting to iteratively construct an additive model,

$$\hat{y}_i^{(t)} = \hat{y}_i^{(t-1)} + f_t(\mathbf{x}_i), \quad (3)$$

where f_t represents the regression tree added at iteration t . The objective function minimized at each step is

$$\mathcal{L}^{(t)} = \sum_{i=1}^n \ell(y_i, \hat{y}_i^{(t)}) + \Omega(f_t), \quad (4)$$

where $\ell(\cdot)$ is a differentiable loss function, such as logistic loss, and $\Omega(\cdot)$ is a regularization term penalizing tree complexity to prevent overfitting.

We further explore neural network-based approaches. The Artificial Neural Network (ANN) considered here is a shallow feed-forward network with a single hidden layer. For hidden representations computed as $\mathbf{h} = \phi(\mathbf{W}_1 \mathbf{x}_i + \mathbf{b}_1)$, where $\phi(\cdot)$ is an activation function such as ReLU, the predicted probability is given by

$$\hat{y}_i = \sigma(\mathbf{w}_2^\top \mathbf{h} + b_2). \quad (5)$$

Finally, we utilize a Deep Neural Network (DNN) with L hidden layers. The prediction is expressed as

$$\hat{y}_i = \sigma(f^{(L)} \circ f^{(L-1)} \circ \dots \circ f^{(1)}(\mathbf{x}_i)), \quad (6)$$

where $f^{(l)}(\cdot)$ represents the non-linear transformation at layer l . To address class imbalance, the DNN is trained using the focal loss function defined as follows

$$\mathcal{L}_{\text{focal}} = - \sum_{i=1}^n \alpha(1 - \hat{y}_i)^\gamma y_i \log \hat{y}_i + (1 - \alpha) \hat{y}_i^\gamma (1 - y_i) \log(1 - \hat{y}_i), \quad (7)$$

where α balances the class weights and γ down-weights well-classified examples, focusing learning on harder cases.

The inclusion of these models provided both breadth and depth in evaluation. Logistic regression offered transparency and interpretability, serving as a benchmark for assessing gains from more complex models. Random forests and XGBoost captured non-linear interactions through ensemble learning, with XGBoost effectively handling imbalance via class weighting. Neural networks represented higher-capacity learners: the ANN provided a shallow deep learning baseline, while the DNN leveraged deeper architectures and focal loss for improved representation learning in imbalanced fraud detection tasks.

To combine the strengths of gradient boosting and deep neural networks, a hybrid ensemble model was constructed using a weighted averaging strategy. In this approach, XGBoost and the DNN were trained independently on the same training data, and their predicted probabilities were combined using a weighted linear combination. Specifically, the final ensemble prediction is given by

$$\hat{y}_i = \lambda \cdot \hat{y}_i^{\text{XGB}} + (1 - \lambda) \cdot \hat{y}_i^{\text{DNN}}, \quad (8)$$

where $\hat{y}_i^{\text{XGB}}$ and $\hat{y}_i^{\text{DNN}}$ denote the predicted probabilities from XGBoost and the DNN respectively, and λ is a weighting parameter.

Algorithm 2: Hybrid Ensemble Training (XGBoost + DNN)

Input: $X_{\text{train}}, y_{\text{train}}, X_{\text{val}}, y_{\text{val}}$

Output: Hybrid model H

Train XGB on $(X_{\text{train}}, y_{\text{train}})$ with imbalance weight w

$xgb_probs \leftarrow \text{XGB.predict}(X_{\text{val}})$

Train DNN independently on $(X_{\text{train}}, y_{\text{train}})$ with focal loss

$dnn_probs \leftarrow \text{DNN.predict}(X_{\text{val}})$

$ensemble_probs \leftarrow 0.6 \cdot xgb_probs + 0.4 \cdot dnn_probs$

Tune threshold on validation set to maximize F1-score

$H \leftarrow (\text{XGB}, \text{DNN}, \text{weights}, \text{threshold})$

return H

In this hybrid approach, XGBoost contributes structured feature learning and interpretability, while the DNN captures non-linear patterns, enabling robust fraud detection in imbalanced and high-dimensional environments. The weights (0.6 for XGBoost, 0.4 for DNN) were selected based on validation performance, giving slightly greater influence to XGBoost due to its stability on tabular data while still leveraging the DNN's capacity to refine decision boundaries. This weighting scheme produced the best balance between recall and precision, ensuring that the ensemble remains both accurate and adaptable for practical deployment.

For further clarification, in Algorithm 2, “tuning the threshold to maximize F1 on (y_{val}) ” refers to selecting an optimal decision threshold that converts predicted probabilities into binary labels. Instead of using a fixed threshold (e.g., 0.5), a range of candidate thresholds is evaluated, and the one that maximizes the F1-score on the validation set is chosen. This step is particularly important in imbalanced settings such as fraud detection, where the default threshold often yields poor recall for the minority class. Optimizing the threshold with respect to the F1-score enables a better balance between precision and recall, improving detection performance. Importantly, threshold tuning is a post-training calibration step rather than part of model learning. Model parameters are trained exclusively on the training set, while the validation set is used only to determine the operating point on the precision–recall curve. No parameter updates are performed on the validation data. To limit potential bias, the validation set is strictly separated from training data, and the selected threshold is fixed before final evaluation. This practice is standard in imbalanced classification and reflects real-world deployment, where decision thresholds are calibrated to meet operational objectives. While more conservative approaches (e.g., nested cross-validation) could further reduce bias, they are computationally expensive at this scale.

3.5. Model architecture and hyperparameter configuration

The Artificial Neural Network (ANN) model consists of two hidden layers with 128 and 64 neurons, respectively. Both hidden layers employ the ReLU activation function, followed by dropout layers with a rate of 0.3 to mitigate overfitting. The output layer contains a single neuron with a sigmoid activation function to produce probabilistic predictions for binary classification. The model is trained using the Adam optimizer with a learning rate of 0.001 and binary cross-entropy loss. Training is conducted for up to 20 epochs with a batch size of 512, and early stopping with a patience of 3 epochs is applied to prevent overfitting.

The Deep Neural Network (DNN) is designed as a deeper architecture with three hidden layers containing 512, 256, and 128 neurons, respectively. Instead of standard ReLU activation, LeakyReLU is employed to mitigate the “dying ReLU” problem and improve gradient propagation in deeper networks. Each hidden layer is followed by batch normalization and dropout with rates of 0.4, 0.4, and 0.3, respectively, to enhance training stability and generalization. The output layer uses a sigmoid activation function to produce probability estimates. The model is trained using the Adam optimizer with a learning rate of 0.001 and focal loss, which is particularly effective in handling class imbalance by emphasizing difficult-to-classify samples. Training is performed for up to 20 epochs with a batch size of 1024, with early stopping and learning rate reduction applied to improve convergence.

The Random Forest model consists of 100 decision trees with a maximum depth of 15. Class imbalance is handled through built-in class weighting, ensuring that minority class samples receive higher importance during training. The model utilizes all available CPU cores to improve computational efficiency. The choice of tree depth and number of estimators reflects a balance between model expressiveness and overfitting control.

Regularization is incorporated across all models to enhance generalization performance. In the neural network models, dropout is used to reduce overfitting by preventing co-adaptation of neurons, while batch normalization stabilizes training and accelerates convergence. Class imbalance is addressed through cost-sensitive learning techniques, including class weighting and focal loss, which adjust the learning process without modifying the original data distribution. Early stopping is applied to prevent overfitting by halting training when validation performance no longer improves.

Regarding activation functions, ReLU is used in the ANN due to its computational efficiency and strong empirical performance. In the deeper DNN architecture, LeakyReLU is adopted to avoid inactive neurons and improve gradient flow. The sigmoid activation function is used exclusively in the output layer, as it is standard for binary classification tasks and enables probabilistic interpretation of predictions.

Hyperparameters were selected based on a combination of prior literature, empirical validation, and computational considerations. While systematic hyperparameter tuning methods such as grid search or Bayesian optimization could potentially yield marginal performance improvements, these approaches are computationally expensive given the scale and dimensionality of the data. Instead, we adopt a pragmatic approach by using well-established configurations and validating them through stratified train-validation splits. Furthermore, techniques such as early stopping and adaptive learning rate scheduling provide implicit tuning during training, allowing the models to converge to suitable parameter settings efficiently.

3.6. Experimental design

The methodological framework discussed in the previous sections laid the foundation for a structured experimental design. Having defined the preprocessing pipeline, imbalance handling strategies, and predictive models, the next step was to implement these methods in a controlled series of experiments. These experiments were carefully structured to test models of varying complexity under realistic fraud detection conditions.

3.6.1. Objectives and rationale

The primary objective of the experiments was to evaluate the effectiveness of various supervised machine learning models for detecting fraud transactions using the IEEE-CIS dataset. Three considerations guided the experimental design. First, simple and interpretable models were assessed as baseline approaches. Second, ensemble methods, including Random Forest and XGBoost, were examined for their ability to improve discrimination under severe class imbalance. Third, deep neural networks and hybrid ensembles were investigated to capture non-linear patterns potentially missed by tree-based methods.

This approach reflects the operational priorities of fraud detection, which require a balance between predictive accuracy, interpretability, scalability, and computational efficiency. The experiments followed a staged workflow: dataset preparation and preprocessing, class imbalance mitigation, evaluation of increasingly complex model families, and performance assessment using metrics appropriate for imbalanced classification. The overall workflow and the range of models are illustrated in Fig. 1.

Scalability and computational efficiency were key factors in both model selection and experimental design, given the large scale and high dimensionality of the data. These considerations influenced not only which models were evaluated, but also how they were configured and compared.

Model selection was guided by the need to balance predictive performance with computational feasibility. Tree-based ensemble models such as Random Forest and XGBoost were chosen because they scale efficiently to large tabular datasets and provide strong performance with relatively low training complexity. In particular, XGBoost incorporates parallelized tree construction and optimized memory usage, making it suitable for high-volume transactional data. Random Forest serves as a computationally efficient baseline, enabling comparison with more advanced models while maintaining stable and scalable training behavior.

Deep learning models, including ANN and DNN, were introduced to capture complex non-linear relationships that may not be fully exploited by tree-based methods. However, these models are computationally more expensive due to iterative gradient-based

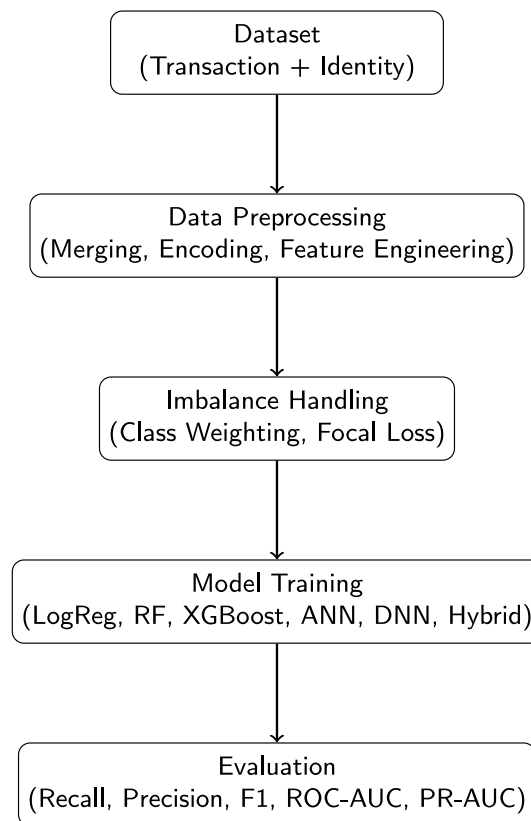


Fig. 1. Staged workflow of the experimental design, from dataset preparation to evaluation.

Table 3

Comparison of performance metrics due to different imbalance handling strategies.

Method	ROC-AUC	PR-AUC	F1-score	Precision	Recall
SMOTE + ensemble	0.9037	0.6068	0.60	0.73	0.51
Class weighting + focal loss (proposed)	0.9638	0.6582	0.74	0.78	0.69

optimization and the requirement for feature scaling. To ensure computational efficiency, the neural network architectures were constrained in depth and size, and training was controlled using early stopping, batch processing, and learning rate scheduling.

In addition to training efficiency, inference time was explicitly evaluated to assess the suitability of each model for real-time deployment. Inference latency was measured as the average prediction time per transaction (milliseconds per transaction) on the validation set. This metric is particularly important in fraud detection systems, where models must process large volumes of transactions with minimal delay. The measured inference times for all models are reported in Table 4.

The hybrid ensemble model was explicitly designed to balance scalability and predictive performance. XGBoost contributes efficient feature learning and fast inference, while the DNN enhances predictive capability through non-linear representation learning. While neural network models incur higher inference cost due to forward propagation through multiple layers, tree-based models such

Table 4

Comparative performance of models on fraud detection task, including external benchmark and inference time.

Model	Accuracy	Recall	Precision	F1-Score	ROC-AUC	PR-AUC	Inference Time (ms/transaction)
Logistic regression (LR)	77%	0.69	0.10	0.18	0.7959	0.1772	0.02
Random forest (RF)	94%	0.68	0.33	0.45	0.9122	0.5856	0.15
XGBoost (XGB)	89%	0.81	0.22	0.35	0.9276	0.6187	0.08
Enhanced XGBoost (XGB-adv)	95%	0.83	0.42	0.55	0.9628	0.7676	0.09
Artificial neural network (ANN)	98%	0.43	0.91	0.59	0.9160	0.6403	0.25
Deep neural network (DNN)	98%	0.51	0.82	0.63	0.9182	0.6582	0.30
XGB + DNN ensemble	98%	0.69	0.78	0.74	0.9638	0.7897	0.35
OLightGBM (Taha and Malebary (2020))	98%	–	0.97	0.5695	0.9288	–	–

as XGBoost provide lower latency through efficient decision-tree traversal. The hybrid ensemble introduces a modest additional overhead by combining predictions from both models, but remains within practical limits for real-time applications.

From an application perspective, these factors are critical. Fraud detection systems must process large volumes of transactions with low latency. The inclusion of inference-time evaluation ensures that the proposed approach is not only accurate but also practical for deployment in latency-sensitive environments.

3.6.2. Models under evaluation

Six models, representing distinct methodological families, were selected to evaluate performance across the comparative framework. Traditional baselines and ensemble learners included Logistic Regression as a transparent, low-cost benchmark; Random Forest, which captures non-linear interactions and handles missing data effectively; and XGBoost, a gradient boosting algorithm designed for high performance on imbalanced data through iterative boosting, regularization, and class weighting. Neural network models comprised a shallow ANN to assess whether modest architectures could detect fraud patterns beyond tree-based methods, and a deeper DNN with additional hidden layers, dropout regularization, and focal loss to enhance representation and handle class imbalance. Both models were trained independently, and their predicted probabilities were combined using a linear weighting scheme. This approach avoids the complexity of stacking while still leveraging the complementary strengths of boosting and deep learning models. This hybrid integration leveraged both structured feature learning and deep representations, aiming to improve predictive performance in real-world fraud detection.

3.6.3. Evaluation framework

Model evaluation in fraud detection requires careful consideration of extreme class imbalance. Overall accuracy was excluded due to its tendency to overstate performance in majority-dominated datasets. Instead, a combination of complementary metrics was employed to assess both detection effectiveness and operational impact. Recall (sensitivity) was prioritized to minimize costly false negatives, while precision ensured that gains in recall did not lead to excessive false alarms. The F1-score provided a balanced measure of this trade-off. Discriminative ability was captured via ROC-AUC, with additional emphasis on PR-AUC, which better reflects performance on the minority (fraud) class. Finally, a cost-sensitive adjustment penalized false negatives more heavily than false positives, aligning evaluation with the financial consequences of undetected fraud. This multi-metric framework ensured models were assessed for both predictive accuracy and practical utility.

4. Results

This section integrates both exploratory data analysis and modeling, providing a unified presentation of findings. Exploratory data analysis is incorporated directly into the results to illustrate how transaction characteristics and feature distributions inform subsequent model performance. The subsections are structured to highlight specific aspects of the data and connect them to fraud detection outcomes.

4.1. Fraud distribution and class imbalance

The first step in characterizing the dataset was to assess the distribution of fraud versus legitimate transactions. Based on the processed dataset used in this study, there are 16,530 fraud transactions out of a total of 472,432 records, corresponding to a fraud proportion of approximately 3.5%. This value was verified directly from the data as

$$\text{Fraud Ratio} = \frac{\sum_{i=1}^n y_i}{n} = \frac{16530}{472432} \approx 0.035.$$

Although this level of imbalance is less extreme than in certain real-world financial datasets, it still presents a significant challenge for classification models. A naive classifier that always predicts the majority (non-fraud) class would achieve approximately 96.5% accuracy, yet it would fail to detect any fraud transactions. This demonstrates that accuracy alone is not an appropriate metric for evaluating fraud detection performance. To address this issue, evaluation metrics such as precision, recall, F1-score, ROC-AUC, and PR-AUC are used throughout this study, as they better reflect performance on the minority class.

4.2. Evaluation of class imbalance handling strategies - SMOTE

Class imbalance is a fundamental challenge in fraud detection, where fraud transactions typically constitute only a small fraction of the dataset. In this study, we primarily address class imbalance through cost-sensitive learning techniques, including class weighting, focal loss, and threshold optimization. To provide a more comprehensive evaluation, we also investigate the effectiveness of the Synthetic Minority Over-sampling Technique (SMOTE). SMOTE generates synthetic minority class samples by interpolating between existing minority instances, thereby increasing the representation of fraud transactions in the training data. SMOTE was applied to the training dataset prior to model fitting. Specifically, a sampling ratio of 0.2 was used, meaning that the minority class was increased to 20% of the majority class. After applying SMOTE, the augmented dataset was used to train both the XGBoost and deep neural network components of the ensemble model, following the same pipeline as the baseline approach.

The results indicate that the proposed cost-sensitive approach significantly outperforms the SMOTE-based model across all evaluation metrics. In particular, the proposed method achieves higher ROC-AUC, PR-AUC, and F1-score, demonstrating better overall

discrimination and balance between precision and recall. For completeness, we note that generative approaches such as GAN-based methods have also been proposed for imbalance handling. However, these methods introduce significant computational complexity and are difficult to stabilize in high-dimensional tabular data. Given these challenges, and the strong performance of cost-sensitive learning observed in our experiments, we do not pursue GAN-based methods in this study. Overall, the experimental results in Table 3 demonstrate that cost-sensitive learning provides a more effective and stable approach to handling class imbalance in the dataset compared to SMOTE, making it a more suitable choice for tabular fraud detection tasks.

4.3. Transaction and product patterns

Fraud activity often manifests through systematic behaviors rather than random noise, making transaction characteristics an important source of discriminative features. In particular, monetary value and product type reveal clear differences between fraud and legitimate transactions, as illustrated in Fig. 2. Fig. 2a presents the distribution of transaction amounts by class. Fraud activity is disproportionately concentrated in very low value transactions, often below 200 USD, which suggests that fraudsters frequently conduct small “test” purchases to confirm the validity of stolen credentials before escalating to larger transactions. At the other extreme, there is also a visible concentration of fraud in very high value purchases, reflecting opportunistic attempts to maximize financial gain once an account has been compromised. Legitimate transactions, by contrast, are more evenly spread across the range but cluster most heavily in the mid value segment, particularly below 500 USD, after which their frequency declines sharply. This divergence between fraud and legitimate spending behaviors highlights the deliberate strategies used by fraudsters to balance concealment and profitability. Fig. 2b shows fraud prevalence across product categories. Fraud rates are highest in category C, followed by category S, whereas categories H, R, and especially W exhibit much lower levels of fraud activity. These discrepancies align with differences in product risk profiles: categories dominated by digital or card not present transactions are more vulnerable to exploitation, while those tied to physical goods or requiring stronger verification demonstrate greater resilience. Incorporating such product level distinctions into predictive models provides highly discriminative signals that enhance the effectiveness of both ensemble and neural network based approaches to fraud detection.

4.4. Temporal and geolocation patterns of fraud

Fraud transactions also exhibit systematic temporal and geographical behaviors rather than occurring uniformly across time and space. These patterns, summarized in Fig. 3, reveal how fraud risk is influenced by daily cycles, weekly rhythms, and regional contexts. Fig. 3a shows fraud distribution across hours of the day. Fraud activity peaks during late-night and early-morning hours, a pattern consistent with reduced customer vigilance and lower institutional monitoring during off-peak periods. Fig. 3b presents the distribution of fraud rates by day of the week. The x-axis represents the seven days (Monday to Sunday), obtained by mapping the transaction timeline into a weekly cycle using modular arithmetic on the transaction day index. This transformation converts the continuous transaction timeline into a categorical representation that captures weekly periodic behavior. The results indicate that fraud rates are relatively stable across weekdays, with a slight decrease observed on Fridays and moderate increases during weekends, particularly on Saturday and Sunday. This pattern may reflect reduced monitoring and lower operational oversight during weekends, as well as increased online transaction activity, which can create more opportunities for fraud behavior. The lower fraud rate on Fridays may be associated with more structured transaction patterns and stronger institutional controls during standard business periods. While temporal variation exists, fraud is not strongly concentrated on a single day but instead reflects broader behavioral and operational dynamics. By explicitly encoding the day-of-week structure, this visualization provides a clearer and more interpretable representation of weekly fraud patterns compared to the original sequential time-based plot. Fig. 3c and d highlight geographical differences. Fraud rates vary significantly across regions and countries, with certain locations showing disproportionately high prevalence. These discrepancies may reflect both differences in fraudster targeting strategies and variability in regional payment

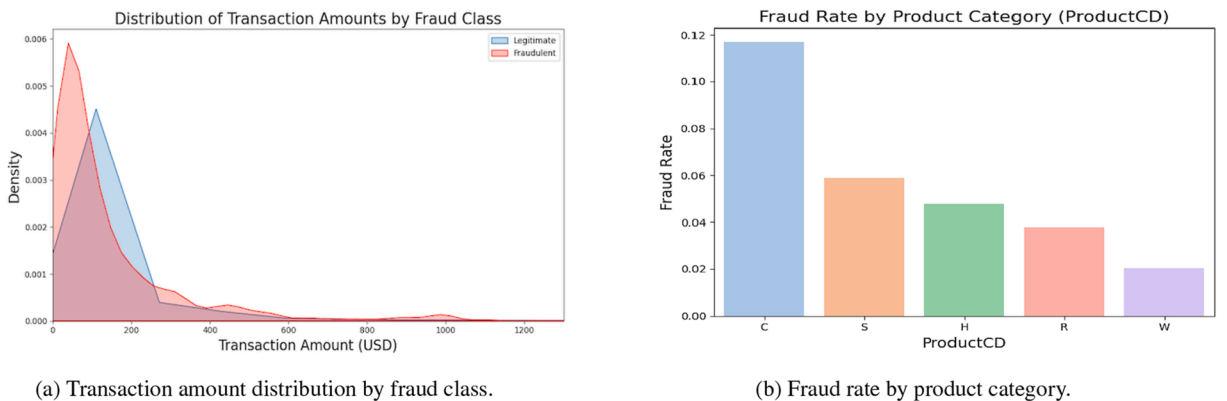
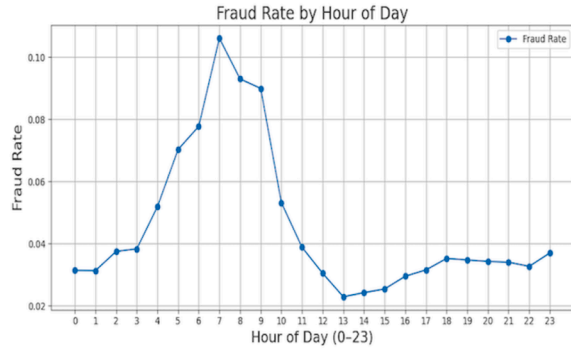
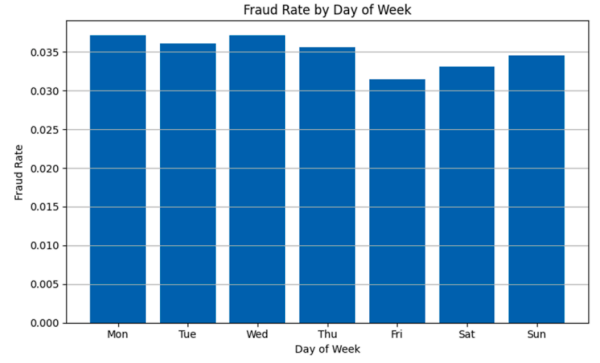


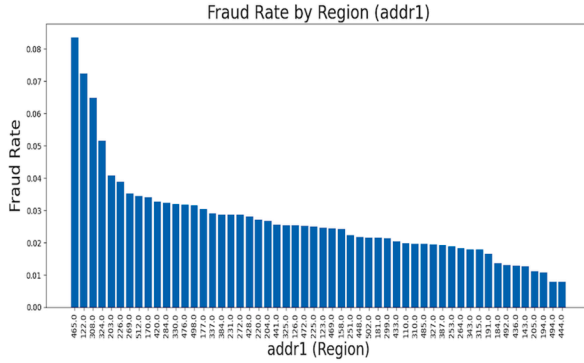
Fig. 2. Fraud patterns across transaction amounts and product categories.



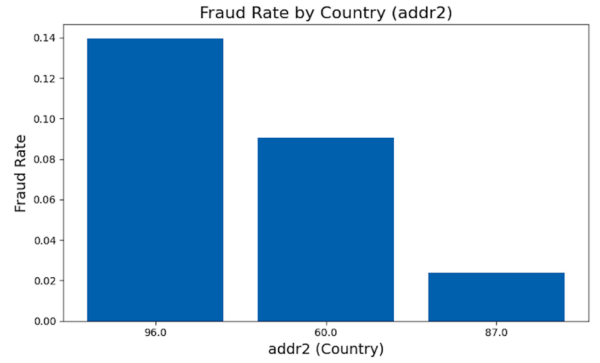
(a) Fraud distribution by hour of day.



(b) Fraud distribution by day of week.



(c) Fraud distribution by region.



(d) Fraud distribution by country.

Fig. 3. Temporal and spatial fraud patterns: (a) hour of day; (b) day of week; (c) region; (d) country.

infrastructures. Such spatial heterogeneity reinforces the value of incorporating geolocation features into fraud detection models, as they provide powerful discriminative signals when combined with transaction and identity attributes.

Temporal and geolocation information are explicitly incorporated into the predictive models through feature engineering and preprocessing, rather than being used solely for exploratory visualization. Temporal information is derived directly from the TransactionDT variable, which represents the elapsed time since a reference point. From this variable, the hour of the day is constructed as:

$$\text{hour_of_day} = \left(\frac{\text{TransactionDT}}{3600} \right) \bmod 24. \quad (9)$$

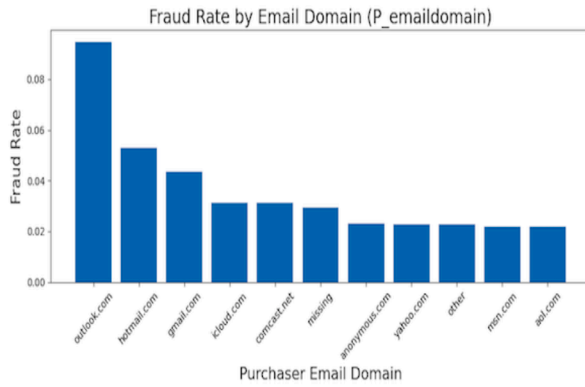
This transformation converts raw timestamps into a cyclical daily feature that captures time-of-day fraud patterns observed during exploratory analysis. The resulting variable is included as a numerical feature in the model input.

Geolocation information is incorporated using the addr1 (region) and addr2 (country) variables provided in the dataset. As these variables are categorical, they are encoded using label encoding during preprocessing, where each unique category is mapped to a numerical value. This representation enables both tree-based models and neural networks to learn location-specific fraud patterns.

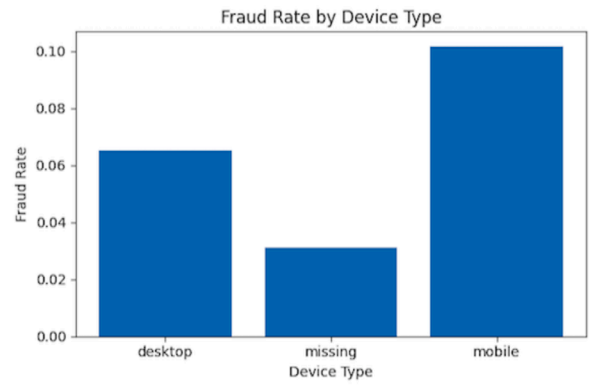
Following feature engineering and encoding, all temporal and geolocation variables are included in the final feature matrix X , which is used to train all models in this study, including Random Forest, XGBoost, ANN, DNN, and the hybrid ensemble. Consequently, the models are able to directly learn from temporal cycles and geographical risk patterns identified in the data. By incorporating these engineered features into the training process, the temporal and spatial patterns discussed in Section 4.3 are not only observed but are actively leveraged to improve fraud detection performance.

4.5. Identity-based feature insights

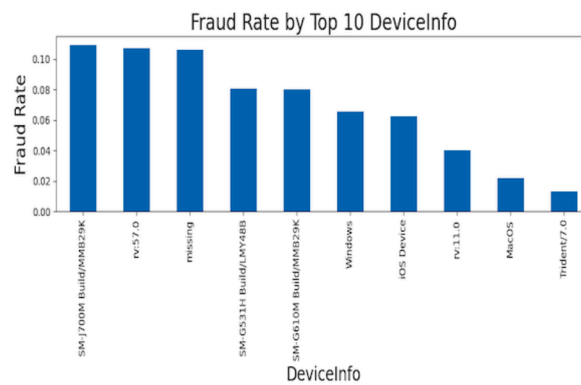
Identity-related features provide some of the strongest discriminative signals in fraud detection, and Fig. 4 highlights three representative patterns. Fig. 4a shows that free email domains are disproportionately associated with fraud activity, whereas institutional or corporate domains exhibit much lower fraud rates, reflecting fraudsters' preference for anonymous or disposable services. Fig. 4b compares fraud rates across device types, with mobile and tablet transactions showing higher fraud levels than desktop, consistent with weaker authentication mechanisms and less reliable device fingerprinting. Fig. 4c illustrates variation by device information, where certain identifiers appear disproportionately in fraud transactions, suggesting the use of emulated or spoofed devices



(a) Fraud distribution by email domain.



(b) Fraud distribution by device type.



(c) Fraud distribution by device information.

Fig. 4. Identity-based fraud patterns: (a) email domain; (b) device type; (c) device information.

or the systematic reuse of compromised profiles. Together, these findings underscore why identity-linked attributes are heavily weighted by both tree-based ensembles and deep learning models in fraud detection.

4.6. Results of model performance

This section presents a comparative evaluation of all seven machine learning models developed for transaction fraud detection: Logistic Regression (LR), Random Forest (RF), XGBoost (XGB), Enhanced XGBoost (XGB-Adv), Artificial Neural Network (ANN), Deep Neural Network (DNN), and the XGBoost + DNN Ensemble. Table 4 summarizes the validation performance of all models.

Metrics include ROC-AUC and PR-AUC for ranking capability, as well as accuracy, precision, recall, and F1-score for fraud detection effectiveness. In addition, inference time (measured in milliseconds per transaction) is reported to assess computational efficiency in real-time deployment scenarios.

The results show a clear performance gap between baseline models and advanced approaches. Logistic Regression suffers from low precision (0.10) and F1-score (0.18), making it unsuitable for fraud detection. Random Forest improves on these metrics, reaching an F1-score of 0.45, but still falls behind gradient boosting methods in capturing complex relationships in the data. Standard XGBoost raises recall to 0.81, boosting fraud capture rates, but precision (0.22) remains low, leading to a higher false-positive rate.

Enhanced XGBoost demonstrates the most significant improvement among single models, delivering the highest recall (0.83) and substantially better precision (0.42) than standard XGBoost. This balance results in a higher F1-score (0.55) and strong PR-AUC (0.7676), reflecting better performance in the imbalanced fraud detection setting. The improvement can be attributed to careful hyperparameter tuning and regularization, which reduce overfitting while improving fraud detection sensitivity. For use cases prioritizing maximum fraud capture with acceptable false positives, Enhanced XGBoost is a strong candidate.

The XGB + DNN Ensemble outperforms all other models in overall balance, achieving the highest F1-score (0.74) and PR-AUC (0.7897), alongside a high recall (0.69) and precision (0.78). By combining Enhanced XGBoost's structured feature learning with the DNN's deep representation capability, the ensemble reduces weaknesses present in each standalone model. This makes it well-suited for real-world usage, where both detecting fraud and minimizing false positives are critical for efficiency and customer trust.

While Table 4 shows that some performance differences (e.g., ROC-AUC of 0.9638 vs. 0.9628) appear small, it is important to determine whether these improvements are statistically meaningful and consistent rather than artifacts of a particular train-validation split. To address this, additional experiments were conducted using multiple random data partitions. Specifically, the dataset was split into training (80%) and validation (20%) sets across three different random seeds, using stratified sampling to preserve the class imbalance. For each split, both the Enhanced XGBoost model and the proposed XGB + DNN Ensemble were trained independently and evaluated on the corresponding validation set.

The evaluation metric used for comparison is ROC-AUC. Results are summarized as mean $\pm$ standard deviation across the three runs. The Enhanced XGBoost model achieved an average ROC-AUC of 0.9223 ± 0.0019 , while the proposed ensemble achieved 0.9274 ± 0.0025 . The ensemble consistently outperformed the baseline model across all splits. To formally assess whether this improvement is statistically significant, a paired t-test was conducted on the ROC-AUC scores obtained from each split. This test is appropriate since both models are evaluated on identical validation partitions, enabling a direct paired comparison. The resulting p-value is 0.0054, which is well below the conventional significance threshold of 0.05, indicating that the improvement is statistically significant and unlikely to be due to random variation.

Furthermore, the small standard deviation observed for both models indicates stable performance across different data splits. This demonstrates that the results are robust and not sensitive to the specific 80/20 partition of the data, despite the relatively small proportion of fraud cases. The overall evaluation procedure follows the training and validation pipeline described in Algorithm 2, extended across multiple randomized splits. Overall, these results demonstrate that the proposed XGB + DNN Ensemble provides a consistent and statistically significant improvement over the baseline model. While more extensive validation strategies such as k-fold cross-validation could further strengthen this analysis, the current findings already provide strong empirical evidence supporting the robustness and reliability of the proposed approach.

OLightGBM, an optimized version of the Light Gradient Boosting Machine (LightGBM), proposed by Taha and Malebary (2020), is incorporated into the main comparative results (Table 4) to provide a unified and consistent benchmark across all models, rather than being presented in a separate subsection. OLightGBM enhances the standard framework through Bayesian-based hyperparameter optimization to automatically tune model parameters and improve predictive performance. LightGBM itself is a gradient boosting algorithm based on decision trees, designed for efficiency and scalability. It employs techniques such as gradient-based one-side sampling (GOSS) and exclusive feature bundling (EFB) to reduce computational complexity while maintaining high predictive accuracy. The OLightGBM approach builds upon this foundation by optimizing hyperparameters governing tree growth and learning dynamics.

The reported results for OLightGBM are taken directly from the original study, where it achieved a ROC-AUC of 0.9288, accuracy of 98.40% (rounded to 98%), precision of 97.34% (rounded to 0.97), and F1-score of 0.5695. As shown in Table 4, both the Enhanced XGBoost and the proposed XGB + DNN Ensemble outperform OLightGBM in terms of ROC-AUC, achieving 0.9628 and 0.9638, respectively. This indicates that the proposed models provide improved discriminative capability relative to a strong published benchmark.

It is important to note that the OLightGBM results are taken from the original publication and may be based on different datasets, preprocessing pipelines, and evaluation protocols. Therefore, this comparison is intended to provide contextual benchmarking rather than a strictly controlled experimental comparison. Nevertheless, the results suggest that the proposed models are competitive with, and in this case outperform, existing state-of-the-art approaches.

In addition to predictive performance, inference time results reported in Table 4 provide important insights into the computational efficiency of each model in real-time deployment scenarios. Tree-based models, particularly XGBoost (0.08 ms/transaction) and Enhanced XGBoost (0.09 ms/transaction), achieve low inference latency due to efficient decision-tree traversal, making them highly suitable for large-scale, low-latency fraud detection systems. Logistic Regression exhibits the lowest latency overall (0.02 ms/transaction), although its predictive performance is substantially weaker.

In contrast, neural network models such as ANN (0.25 ms/transaction) and DNN (0.30 ms/transaction) incur higher inference times due to forward propagation through multiple layers. The hybrid XGB + DNN ensemble introduces additional computational overhead by combining predictions from both models, resulting in the highest latency (0.35 ms/transaction) among the evaluated approaches.

These results highlight a clear trade-off between predictive performance and computational efficiency. While the proposed ensemble achieves the best overall detection performance, its higher latency may limit its applicability in ultra-low-latency environments. In such cases, XGBoost provides a strong alternative, offering competitive predictive performance with significantly lower inference time, making it particularly attractive for real-time fraud detection systems where rapid decision-making is critical.

4.7. Comparison with sequential model (LSTM)

To evaluate whether sequence-based modeling improves fraud detection performance, we implemented a Long Short-Term Memory (LSTM) network as a representative state-of-the-art temporal model. Fraud detection is often considered inherently sequential, as user behavior over time may reveal anomalous patterns. Therefore, incorporating an LSTM provides a meaningful benchmark to assess whether modeling temporal dependencies yields performance gains for this dataset.

The LSTM model was trained using sliding-window sequences constructed from the transaction data, with each sequence representing a short temporal context of transactions. Class imbalance was addressed using a weighted binary cross-entropy loss function to ensure that the model does not trivially predict the majority class. However, experimental results demonstrate that the LSTM performs substantially worse than the proposed hybrid ensemble model across all evaluation metrics (as shown in Table 5). Specifically, the LSTM achieves a ROC-AUC of 0.502, PR-AUC of 0.036, and F1-score of 0.013, indicating near-random classification behavior. In

contrast, the proposed XGBoost + DNN ensemble achieves substantially higher performance, with a ROC-AUC of 0.9638, PR-AUC of 0.6582, and F1-score of 0.74 (see Table 6).

The poor performance of the LSTM can be attributed to the structural characteristics of the IEEE-CIS dataset. Although the dataset includes a temporal feature (TransactionDT), it does not provide consistent user-level sequences or long-term behavioral histories. Transactions are not reliably grouped by individual users, and temporal ordering alone does not guarantee meaningful sequential dependencies. As a result, the constructed sequences lack coherent behavioral patterns, limiting the ability of sequence-based models to learn discriminative temporal features. In contrast, the proposed hybrid ensemble model is better aligned with the tabular and high-dimensional nature of the dataset. XGBoost effectively captures complex feature interactions, handles missing values, and is robust to noisy and sparse features commonly found in fraud detection data. The DNN component complements this by learning nonlinear representations and capturing higher-order feature interactions. By combining these two models, the ensemble leverages complementary strengths, resulting in improved generalization and significantly higher predictive performance.

These findings highlight an important insight: model effectiveness is strongly dependent on the underlying data structure. While sequence-based models such as LSTM are powerful for tasks with well-defined temporal dependencies, they are less suitable for tabular fraud detection datasets with weak or irregular sequential patterns. In such cases, hybrid ensemble approaches provide a more effective and practical solution.

4.8. Comparison with kaggle state-of-the-art

To further evaluate the effectiveness of the proposed model, we compare our results with top-performing solutions from the IEEE-CIS Kaggle fraud detection competition.

According to the 1st place solution of the IEEE-CIS Kaggle competition (Deotte et al., 2019), the final model was an ensemble of gradient boosting algorithms, including XGBoost, LightGBM, and CatBoost, combined with extensive feature engineering, validation strategies, and stacking techniques. This approach achieved a private leaderboard ROC-AUC of approximately 0.9459 and a public leaderboard score of 0.9677. Similarly, the 5th place solution (H. M. et al., 2019) relied on an ensemble of LightGBM models with user-level feature aggregation and achieved a private leaderboard score of 0.9425. These results demonstrate that tree-based ensemble methods are consistently effective for this dataset.

In comparison, the proposed hybrid ensemble achieves a ROC-AUC of 0.9638 and F1-score of 0.74, demonstrating competitive performance relative to these state-of-the-art solutions. It is important to note that direct comparison with Kaggle leader board results is not strictly equivalent, as competition settings involve hidden test sets and more stringent validation protocols. Nevertheless, the results indicate that the proposed model achieves strong predictive performance within a standard experimental framework. Unlike the Kaggle-winning solutions, which rely heavily on complex feature engineering and dataset-specific techniques, the proposed method emphasizes a principled integration of tree-based learning and deep neural networks. This results in a simpler, more reproducible modeling pipeline while maintaining strong predictive performance and interpretability through SHAP-based analysis. Interestingly, the observations reported in the Kaggle winning solution further support our findings.

4.9. ROC and Precision–Recall curve analysis

This section compares four models to evaluate how the proposed approaches perform against strong baselines. Random Forest is included as a representative of traditional machine learning methods, while the Deep Neural Network represents deep learning approaches. These are compared with the Enhanced XGBoost (XGB-Adv) and the XGB + DNN Ensemble, which combine boosting and neural networks. This comparison highlights the improvements of the proposed methods over both classical and deep learning models. Fig. 6 presents the ROC and Precision–Recall (PR) curves for the four models. Both plots confirm the superior performance of the proposed methods over the baselines. In Fig. 5a, the XGB + DNN Ensemble achieved the highest ROC-AUC of 0.9638, followed closely by XGB-Adv at 0.9628. Among the baselines, the DNN (0.9182) slightly outperformed the RF (0.9122). The steep initial slope and proximity of the proposed models' curves to the top-left corner illustrate their strong discriminative ability across thresholds. Fig. 5b shows the PR curves, which are more informative for imbalanced datasets. The XGB + DNN Ensemble again outperformed all models

Table 5

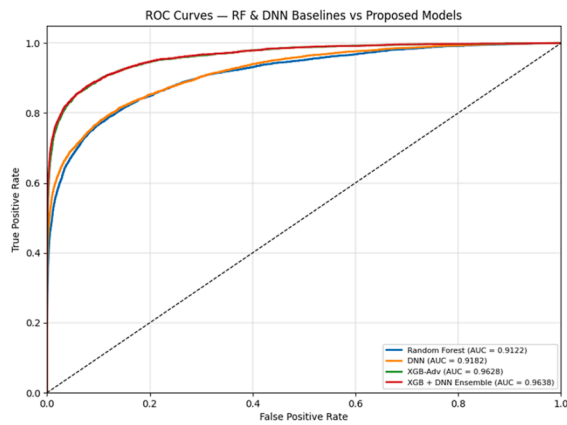
Performance comparison between LSTM and the proposed ensemble model.

Model	ROC-AUC	PR-AUC	F1-score	Precision	Recall
LSTM	0.502	0.036	0.013	0.030	0.008
XGB + DNN ensemble	0.9638	0.6582	0.74	0.78	0.69

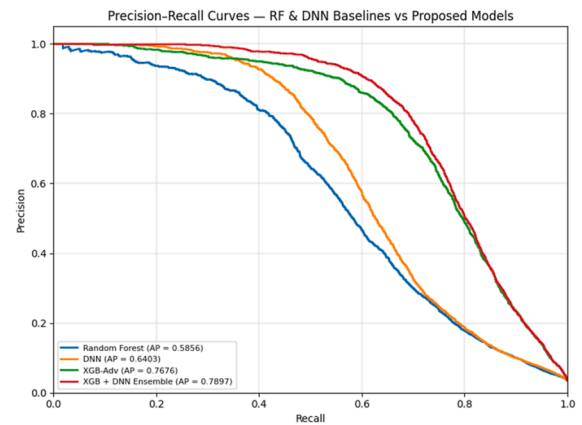
Table 6

Comparison with Kaggle top solutions.

Method	ROC-AUC	PR-AUC	F1-score
Kaggle 1st place ensemble [46]	0.94–0.95	–	–
Kaggle 5th place LightGBM ensemble [47]	0.94	–	–
Proposed XGB + DNN ensemble	0.9638	0.6582	0.74



(a) ROC curves for RF, DNN, XGB-Adv, and XGB + DNN Ensemble.



(b) PR curves for RF, DNN, XGB-Adv, and XGB + DNN Ensemble.

Fig. 5. ROC and Precision–Recall curve analysis for baseline and proposed models: (a) ROC curves; (b) PR curves.

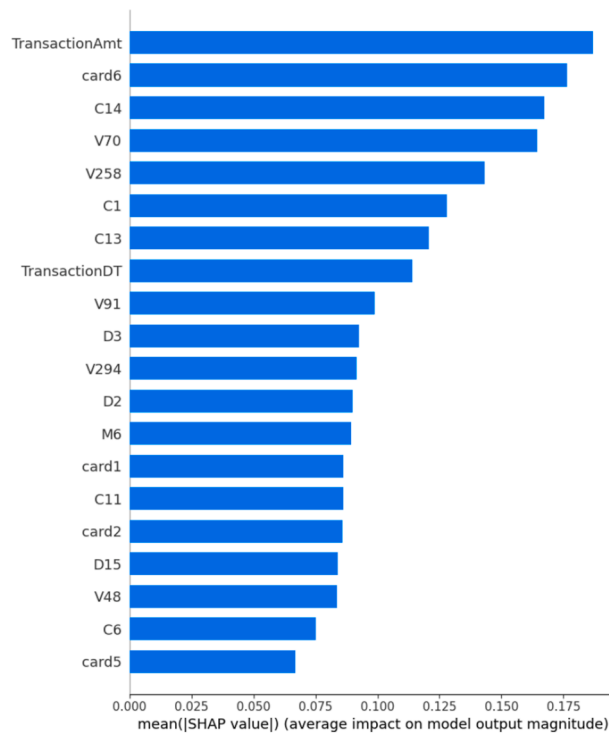


Fig. 6. Global feature importance—bar plot.

with an Average Precision (AP) of 0.7897, followed by XGB-Adv at 0.7676. Among the baselines, the DNN achieved 0.6403, outperforming the RF at 0.5856. The proposed models maintain higher precision at varying recall levels, demonstrating superior handling of the minority fraud class.

The results demonstrate that while traditional models such as Logistic Regression and Random Forest provide useful baselines, they fall short under severe class imbalance. Enhanced XGBoost achieved substantial gains, and the hybrid XGB + DNN Ensemble consistently outperformed all other approaches, delivering the highest recall, F1-score, ROC-AUC, and PR-AUC. Exploratory analyses further revealed clear fraud patterns across transaction amounts, product categories, temporal cycles, and identity-related features, confirming their importance as discriminative signals. Taken together, these findings underscore both the necessity of advanced ensemble methods and the value of incorporating domain-specific patterns, providing a comprehensive foundation for fraud detection pipelines.

4.10. Results on feature interpretability - SHAP based feature analysis

This section extends the analysis beyond performance metrics to interpret the results and their practical significance. While the previous section identified the best-performing models under class imbalance, the focus here is on understanding the mechanisms driving their success, the insights provided by key features, and their implications for real-world fraud detection. Interpretability is particularly critical in financial applications, where transparency and accountability are required for deployment. To this end, SHAP values are computed for the Enhanced XGBoost model, the best-performing single learner, to quantify both global and local feature contributions. The analysis is conducted from two complementary perspectives: (i) global feature importance, using mean absolute SHAP values to identify the most influential predictors across all transactions, and (ii) local interpretation, using SHAP beeswarm plots to examine how feature values, their direction, and distribution influence fraud predictions at the instance level.

4.10.1. Global feature importance in fraud context

The global SHAP importance plot (Fig. 6) shows that *TransactionAmt* is the most influential feature, followed by identity-related features such as *card6*, *C14*, *C1*, and *C13*, as well as behavioral variables including *V70* and *V258*.

This ranking is consistent with prior fraud detection studies, where transaction amount and identity consistency are primary indicators of fraud. Fraud transactions often occur at extreme values: low-value transactions are used as “test” transactions to validate stolen credentials, while high-value transactions reflect attempts to maximize financial gain. Identity-related features capture deviations from normal user behavior. In legitimate transactions, these features tend to be stable, whereas fraud introduces anomalies such as unusual transaction counts or inconsistent card usage. The importance of these features confirms that fraud detection relies heavily on identifying such deviations.

4.10.2. Local SHAP interpretation and feature effects

The SHAP beeswarm plot shown in Fig. 7 provides detailed transaction-level insights by combining feature values with their contributions to model predictions. In the beeswarm plot, the horizontal axis represents SHAP values, indicating the magnitude and direction of each feature's contribution. Positive SHAP values increase fraud likelihood, while negative values decrease it. The color gradient represents feature values (red = high, blue = low), allowing us to infer whether high or low values of a feature are associated with fraud.

For *TransactionAmt*, high values (red points) are predominantly associated with positive SHAP values, indicating increased fraud risk. However, some low values also contribute positively, suggesting that fraud detection captures both high-value exploitation and low-value probing behavior. This demonstrates that fraud risk cannot be explained by simple thresholds but depends on contextual

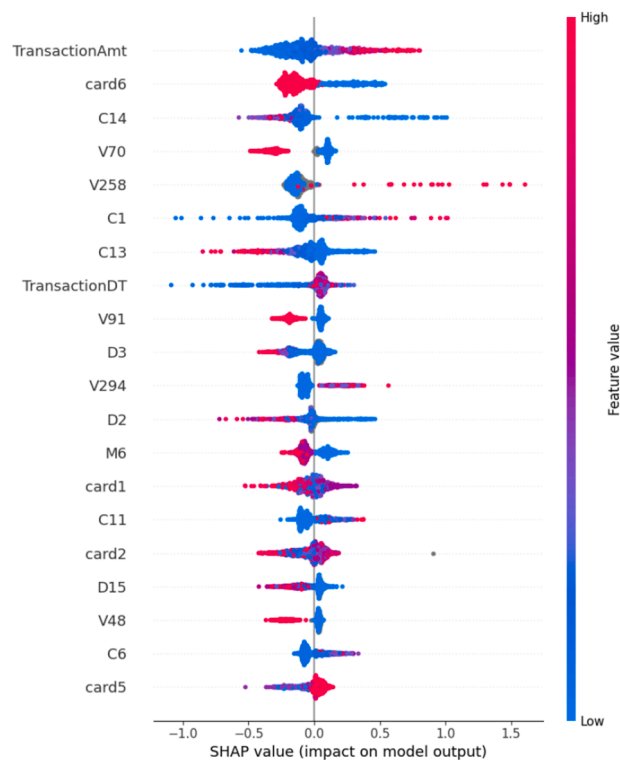


Fig. 7. Local explanation–beeswarm plot.

interactions. For *card6*, distinct clusters of SHAP values indicate that specific categories consistently increase fraud probability. This suggests that certain card types or transaction channels are more vulnerable, likely due to differences in authentication mechanisms. Features such as *C14* and *C13* exhibit asymmetric distributions, where extreme values correspond to strong positive SHAP contributions. This indicates that deviations from normal behavioral patterns significantly increase fraud likelihood, providing evidence of anomaly-based detection. The *V-series* features (e.g., *V70*, *V258*) show wide and overlapping SHAP distributions, indicating strong interaction effects. These features do not act independently but contribute differently depending on the context of other variables. This highlights the importance of models capable of capturing non-linear relationships.

4.10.3. Comparison with random forest

For comparison, the Random Forest feature importance plot (See Fig. 8) shows a broader distribution of influential variables rather than dominance by a small subset. Features such as *C14* and *C13* emerge as the strongest contributors, while transaction-based attributes like *TransactionAmt*, *V258*, and *V265*, along with several *V-series* variables, also rank highly. This balance indicates that Random Forest relies on both identity signals and behavioral transaction features, capturing multiple dimensions of fraud risk. Importances are more evenly spread across categories, suggesting that no single variable alone drives the predictions but rather a combination of complementary signals.

While this confirms the relevance of key features, Random Forest importances provide only relative weights and lack the ability to explain whether a feature increases or decreases the likelihood of fraud. They also do not account for interactions between features or variation across individual cases. These limitations highlight why more interpretable methods such as SHAP are better suited for high-stakes fraud detection, where fine-grained reasoning and case-level explanations are essential for operational use.

4.10.4. Consistency with prior literature and novel insights

The SHAP analysis demonstrates strong agreement with prior research in financial fraud detection. Previous studies have identified transaction amount, identity consistency, and behavioral irregularities as key indicators of fraud. The prominence of these features in our analysis confirms that the proposed models capture established fraud patterns.

At the same time, the results provide additional insights beyond prior work. In particular, the analysis highlights the importance of anonymized behavioral features (*V-series*), which encode latent interaction patterns not directly observable in raw data. These features exhibit context-dependent contributions, indicating that fraud detection is driven by interactions rather than isolated feature effects. This observation aligns with recent work by (Lin and Gao, 2022), who demonstrate that SHAP-based methods can reveal complex relationships between financial features and improve interpretability in fraud detection models.

4.10.5. Practical implications

The enhanced SHAP interpretation provides actionable insights for real-world fraud detection systems. Financial institutions can leverage these findings to combine rule-based detection (e.g., abnormal transaction amounts or identity inconsistencies) with machine learning models capable of capturing complex feature interactions. Furthermore, SHAP-based explanations improve transparency by providing case-level justifications for predictions. This is critical for regulatory compliance and operational trust in financial applications.

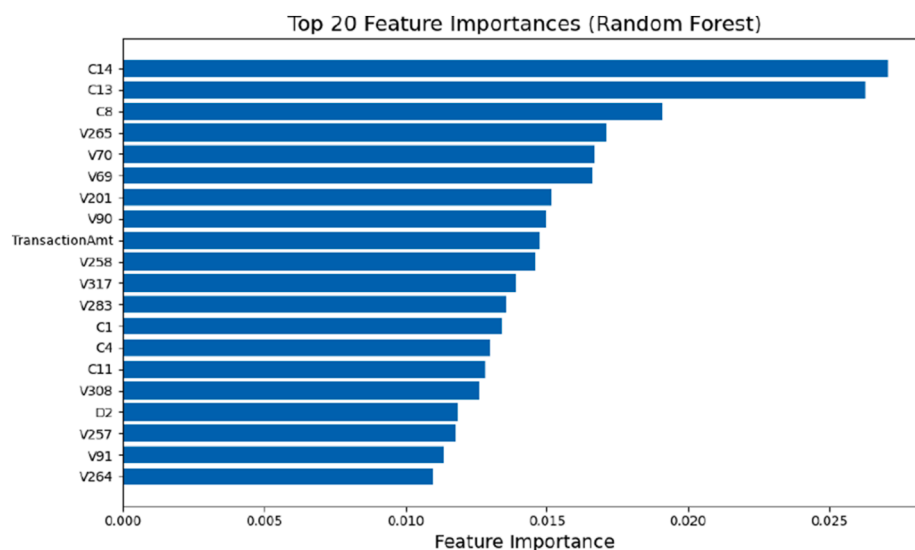


Fig. 8. Random forest - top 20 feature importances.

5. Discussions

The results carry direct implications for real-world fraud detection systems, where predictive performance must be balanced with operational considerations. Model selection cannot rely solely on ROC-AUC or PR-AUC values; institutions must also consider interpretability, latency, and the trade-off between false positives and false negatives.

While this study demonstrates the effectiveness of advanced ensemble models for fraud detection, several limitations should be acknowledged. First, the dataset, although large and representative of real-world payment transactions, is anonymized and lacks certain contextual features (e.g., merchant category codes, customer demographics, and real-time session information). The absence of these variables restricts the interpretability of fraud patterns and may limit the generalizability of the findings across different financial institutions and geographies.

Second, the evaluation was conducted in an offline validation setting. Although metrics such as ROC-AUC and PR-AUC are informative, they do not capture all operational trade-offs. Real-world deployment involves latency constraints, streaming data pipelines, and integration with fraud investigation teams. These aspects were not simulated in this study, and future research should extend validation to production-like environments, including stress testing under high transaction volumes and adversarial attack scenarios.

Third, while SHAP provided valuable explainability for tree-based models, interpretability for deep neural networks remains less developed. Methods such as LIME or integrated gradients offer partial insights, but their stability and regulatory acceptance are still evolving. This creates challenges for deploying complex models in strictly regulated financial domains where transparency is a non-negotiable requirement.

A key limitation of this study is that the IEEE-CIS dataset represents a fixed historical snapshot of transaction data. In real-world fraud detection systems, transaction patterns and fraud strategies evolve continuously over time, a phenomenon commonly referred to as *concept drift*. This occurs when the underlying data distribution or the relationship between features and the target variable changes, often due to adaptive behavior by fraudsters. As fraud detection models are trained on historical data, their performance may degrade when deployed in a live environment if the statistical properties of incoming transactions differ from those observed during training. For example, fraudsters may shift from low-value probing transactions to more sophisticated attacks involving device spoofing or identity obfuscation, rendering previously learned patterns less effective.

While the models proposed in this study demonstrate strong performance under a static evaluation setting, maintaining their effectiveness in practice would require continuous monitoring and adaptation. This may involve periodic retraining on recent data, online learning strategies, or drift detection mechanisms that trigger model updates when significant distributional changes are observed. Furthermore, ensemble approaches such as the proposed XGBoost + DNN model may offer some robustness to moderate drift due to their ability to capture complementary patterns. However, they are not inherently immune to concept drift, and their long-term performance depends on timely updates and data refresh cycles.

6. Conclusion

This study addressed the persistent challenge of fraud detection in highly imbalanced financial transaction data by systematically comparing traditional, ensemble-based, and deep learning approaches. Through rigorous experimentation, it was demonstrated that classical baselines such as Logistic Regression and Random Forest, while interpretable, fail to provide the precision and recall balance required for real-world applications. Enhanced XGBoost improved performance by leveraging gradient boosting with advanced handling of imbalance, yet it was the proposed hybrid XGB + DNN Ensemble that delivered the most effective and robust results across all evaluation metrics. Beyond raw performance scores, the study emphasized the importance of interpretability and operational readiness. SHAP-based feature attribution highlighted the central role of transaction amounts, identity-linked attributes, and temporal patterns in detecting fraud behavior, underscoring the need for multi-dimensional representations of customer activity.

The hybrid XGB + DNN Ensemble, embedded within a structured and adaptive framework, represents a powerful and practical solution for modern fraud detection. By aligning methodological innovation with operational realities, this study demonstrates that effective fraud detection requires not only strong algorithms but also transparent, adaptive, and institutionally aligned systems capable of withstanding the evolving tactics of financial fraud. Together, these findings advance both academic understanding and practical implementation of fraud detection systems. Nevertheless, limitations remain. The anonymized dataset constrained interpretability of certain fraud patterns, and evaluation was restricted to offline validation. Addressing these gaps through richer feature sets, real-time testing, and advanced learning paradigms such as graph-based or federated learning offers promising directions for future research.

CRedit authorship contribution statement

Nguyen Duy Anh Luong: Writing – review & editing, Writing – original draft, Visualization, Validation, Software, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **Shengkun Xie:** Writing – review & editing, Writing – original draft, Supervision, Methodology, Funding acquisition, Formal analysis, Conceptualization.

Grant information

This work is funded by Natural Sciences and Engineering Research Council of Canada Discovery Grant.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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